Scientific and Academic Curriculum

Luca Ghislotti

ISTITUTO NAZIONALE DI FISICA NUCLEARE (INFN), SEZIONE DI PAVIA

UNIVERSITÀ DEGLI STUDI DI BERGAMO

Department of Engineering and Applied Sciences

Microelectronics Laboratory

Bio



Luca Ghislotti is a Computer Engineer and Research Fellow at the Istituto Nazionale di Fisica Nucleare (INFN, National Institute for Nuclear Physics), Sezione di Pavia, working within the CSN2 particle physics division. He obtained his Ph.D. in Engineering and Applied Sciences (Microelectronics for High-Energy Physics) at the Microelectronics Laboratory, Department of Engineering and Applied Sciences, University of Bergamo.

His research focuses on the design and characterization of low-noise, low-power analog integrated circuits for the readout of silicon detectors used in high-energy physics and astrophysics experiments.

He is currently contributing to the General AntiParticle Spectrometer (GAPS) Antarctic balloon experiment as part of the GAPS collaboration. GAPS is a NASA-, JAXA-, INFN-, and ASI-funded project designed to precisely measure the flux of low-energy cosmic-ray antideuterons, antiprotons, and antihelium as an indirect signature of dark matter.

Personal Details

Academic Email Address:	luca.ghislotti@unibg.it
GitHub:	https://github.com/lucaghislo
LinkedIn:	https://www.linkedin.com/in/luca-ghislotti/
Personal Web Page:	https://lucaghislotti.com
Laboratory Web Page:	https://microlab.unibg.it/

Employment and Research Positions

Research Fellow

Istituto Nazionale di Fisica Nucleare (INFN), Sezione di Pavia, Italy (December 2025 – December 2026)

Grant:	INFN Research Fellowship, Call No. 28012/2025
Selection Score:	200/200 (60/60 CV evaluation, 140/140 interview)
Research Title:	"Design and characterization of an analog channel in 65 nm CMOS technology for the readout of the Si(Li) detectors of the GAPS experiment"

Research Associate II

Astrophysics Laboratory, Columbia University, New York (August 1, 2023 – July 31, 2024)

Academic Education and Training

Doctor of Philosophy (Ph.D.) in Engineering and Applied Sciences (XXXVIII cycle)

SSD: IINF-01/A – Electronics (Microelectronics for High Energy Physics)
Università degli Studi di Bergamo, Bergamo (2022 – 2026)

Thesis:	"Design, Testing and Calibration of the GAPS Experiment Si(Li) Tracker Readout ASIC: From the First Flight Campaign Toward the Second Mission Upgrade"
Supervisor:	Prof. Massimo Manghisoni
Co-Supervisor:	Dr. Elisa Riceputi
Ph.D. Coordinator:	Prof. Alessandra Marini
Keywords:	ASIC, Silicon Detectors, Readout Electronics, CMOS Technology, Dark Matter
Defense Date:	January 28, 2026
Evaluation:	Excellent
Link:	https://hdl.handle.net/10446/319848

Master of Science (M.Sc.) in Computer Engineering (LM-32)

Università degli Studi di Bergamo, Bergamo (2020 – 2022)

Thesis:	"Characterization of the readout electronics of the Si(Li) tracker for the first flight of the GAPS experiment" [PDF]
Supervisor:	Prof. Massimo Manghisoni
Keywords:	GAPS, Dark Matter, Spectrometer, Si(Li), ASIC
Final Grade:	110/110 cum laude

Link: https://hdl.handle.net/10446/303265

Bachelor of Science (B.Sc.) in Computer Engineering (L-8)

Università degli Studi di Bergamo, Bergamo (2017 – 2020)

Thesis: "Data Anonymization Techniques: Implementation in the Apache Spark Environment"
Supervisor: Prof. Stefano Paraboschi
Keywords: Data Anonymization, Apache Spark, Apache Hadoop, Privacy, Computer Security
Final Grade: 107/110

Secondary School Diploma in IT and Telecommunications Engineering

Centro Studi Leonardo da Vinci, Bergamo (2012 – 2017)

Final Project: "Where's my car? Sistema di localizzazione GPS" (GPS tracking system)
Final Grade: 100/100

Scholarships and Certificates

Ph.D. Scholarship: Doctoral Fellowship in Engineering and Applied Sciences (Cycle XXXVIII), Università degli Studi di Bergamo
Years: 2022–2025

TOP 10 Student Program: Università degli Studi di Bergamo. Fee exemption award issued to the best students.
Years: 2017–2018, 2020–2021, 2021–2022

Excellence Award (Io e Lode): Confindustria Bergamo. Award for outstanding upper secondary school students.
Years: 2013–2017

FCE (B2 First Certificate): Cambridge Assessment English, Cambridge
Certified in 2017

Memberships

INFN Member: Istituto Nazionale di Fisica Nucleare, CSN2 group (Technological Ph.D., Pavia section)
Years: 2022–2026

IEEE Member: Institute of Electrical and Electronics Engineers (IEEE)
Years: 2023–2026

NPSS Member: IEEE Nuclear and Plasma Sciences Society (NPSS)
Years: 2023–2026

SIE Member: Società Italiana di Elettronica
Years: 2023–2026

Research Profiles

ORCID: orcid.org/0000-0002-7084-5979
Scopus: scopus.com/authid/detail.uri?authorId=57939698700
Web of Science: webofscience.com/wos/author/record/JUU-2100-2023
ADS: ui.adsabs.harvard.edu/search/q=author%3A%22Ghislotti%2C%20Luca
Semantic Scholar: semanticscholar.org/author/L.-Ghislotti/2188430723
Google Scholar: scholar.google.com/citations?user=Xt1_bCYAAAAJ
ResearchGate: researchgate.net/profile/Luca_Ghislotti
INSPIRE-HEP: inspirehep.net/authors/3220693

Scientific Activity

The research activity of Luca Ghislotti focuses mainly on the design of low-noise, low-power analog front-end integrated circuits for semiconductor detector readout in high-energy physics, and their experimental characterization. Since 2022 he has been a member of the GAPS collaboration with affiliations to the University of Bergamo and INFN Pavia (https://gaps1.astro.ucla.edu/gaps/people.php).
The main research topics to date include:

1. **65 nm CMOS readout ASIC for the second GAPS Si(Li) tracker flight**
   Design of a new Application-Specific Integrated Circuit (ASIC) for the second long-duration balloon flight of the GAPS experiment. The chip is an evolution of the SLIDER32 ASIC (180 nm CMOS) used in the first flight, migrating to a 65 nm CMOS technology to achieve improved noise, dynamic range, and integration density.

2. **Characterization and calibration of the Si(Li) tracker electronics**
   Validation of the flight-ready readout electronics of the GAPS Si(Li) tracker, with calibration campaigns focused on ultra-low noise performance and accurate detection of high-energy particles. These activities supported the integration of the instrument for the Antarctic long-duration balloon campaign of late 2024.

3. **Radiation-hard bandgap references**
   Characterization of bandgap voltage references for high-radiation environments, including devices developed for the ARCADIA project and a 28 nm CMOS bandgap for high-energy physics applications.
   Related work: G. Traversi, D. Falchieri, L. Gaioni, M. Manghisoni, M. Pezzoli, L. Ratti, V. Re, L. Ghislotti, *"A radiation hard bandgap voltage reference for the ARCADIA project"*, TWEPP 2022, Bergen, Norway.

4. **DSSC bare module testing for the European XFEL**
   Characterization of bare detector modules consisting of pixelated sensors bump-bonded to readout ASICs, within the DSSC (DEPFET Sensors with Signal Compression) project for the European XFEL in Hamburg. The aim is the development of a MiniSDD-based 1-Megapixel camera for photon science in the 0.25 keV to 6 keV energy range.

   Related contracts:

   - "Validation of the sensors of 10 Large Format Bare Modules for the assembly of the second DSSC camera of the XFEL experiment", within the research project *"Large format x-ray imager with mega-frame readout capability"* (April 2025).
   - "Validation of the sensors of 15 Large Format Bare Modules for the assembly of the second DSSC camera of the XFEL experiment", within the research project *"Large format x-ray imager with mega-frame readout capability"*, specifically focused on the measurement of the back-side current of DEPFET sensors as a function of the applied bias voltage (October 2025).

5. **Wearable device for medical applications**
   Development of a microcontroller-based wearable system for monitoring arteriovenous fistulas in dialysis patients, in collaboration with Istituto di Ricerche Farmacologiche Mario Negri IRCCS.

6. **Gas sensors for air quality monitoring**
   Analysis and characterization of gas sensors for the development of artificial intelligence algorithms able to recognize complex odor signatures and monitor air quality, in collaboration with the University of Brescia.

Journal Articles

Peer-reviewed journal publications in the fields of microelectronics, detector instrumentation, and high-energy physics applications.

[J1] K. Aoyama et al., "The General Antiparticle Spectrometer (GAPS) Antarctic balloon payload," *Nucl. Instrum. Methods Phys. Res. Sect. Accel. Spectrometers Detect. Assoc. Equip.*, vol. 1095, p. 172 167, Mar. 2027, ISSN: 0168-9002. DOI: 10.1016/j.nima.2026.172167

[J2] L. Ghislotti, P. Lazzaroni, M. Manghisoni, V. Re, E. Riceputi, and L. Ratti, "A 65-nm CMOS readout channel based on a charge amplifier with DTMOS feedback for dynamic signal compression," *IEEE Trans. Nucl. Sci.*, vol. 73, no. 7, pp. 2664–2675, Jul. 2026, ISSN: 1558-1578. DOI: 10.1109/TNS.2026.3700799

[J3] M. Manghisoni, L. Ghislotti, P. Lazzaroni, L. Ratti, V. Re, and E. Riceputi, "Modeling Temperature Effects on MOS Capacitors in Charge Amplifier Feedback," *IEEE Trans. Electron Devices*, vol. 73, no. 10, pp. 6207–6214, Oct. 2026, ISSN: 1557-9646. DOI: 10.1109/TED.2026.3724517

[J4] L. Volpicelli and on behalf of the GAPS Collaboration, "New indirect searches of dark matter with the GAPS experiment," *Il Nuovo Cimento C*, vol. 49, no. 1-2, pp. 1–6, Jan. 2026, ISSN: 03905551, 03905551. DOI: 10.1393/ncc/i2026-26006-7

[J5] L. Ghislotti, P. Lazzaroni, M. Manghisoni, and E. Riceputi, *Design of the ANTARES4 readout ASIC for the second flight of the GAPS experiment: Motivations and requirements*, Oct. 2025. DOI: 10.20944/preprints202510.0181.v1 Preprints: 2025100181.

[J6] L. Ghislotti, P. Lazzaroni, M. Manghisoni, and E. Riceputi, "Design of the ANTARES4 readout ASIC for the second flight of the GAPS experiment: Motivations and requirements," *Particles*, vol. 8, no. 4, p. 89, Dec. 2025, ISSN: 2571-712X. DOI: 10.3390/particles8040089

[J7] M. Manghisoni, L. Ghislotti, P. Lazzaroni, V. Re, E. Riceputi, L. Ratti, L. Fabris, M. Boezio, G. Zampa, and M. Xiao, "X-ray and particle detection with the si(li) tracker module of the GAPS experiment," *IEEE Trans. Nucl. Sci.*, pp. 1–1, 2025, ISSN: 1558-1578. DOI: 10.1109/TNS.2025.3616408

[J8] G. Traversi, R. Ballabriga, D. Ceresa, L. Gaioni, L. Ghislotti, S. Michelis, and G. Wegrzyn, "A 1 grad TID-tolerant bandgap voltage reference for HEP applications in 28 nm CMOS node," *IEEE Trans. Nucl. Sci.*, pp. 1–1, 2025, ISSN: 1558-1578. DOI: 10.1109/TNS.2025.3649520

[J9] S. Botticini et al., "Index air quality monitoring for light and active mobility," *Sensors*, vol. 24, no. 10, p. 3170, Jan. 2024, ISSN: 1424-8220. DOI: 10.3390/s24103170

[J10] L. Ghislotti, M. Boezio, L. Fabris, P. Lazzaroni, M. Manghisoni, L. Ratti, V. Re, E. Riceputi, and G. Zampa, "Energy threshold calibration of the GAPS experiment Si(Li) tracker readout electronics," *Il Nuovo Cimento C*, vol. 47, no. 3, pp. 1–5, Apr. 2024, ISSN: 03905551, 03905551. DOI: 10.1393/ncc/i2024-24121-1

[J11] M. Manghisoni, L. Ghislotti, P. Lazzaroni, V. Re, E. Riceputi, L. Ratti, L. Fabris, M. Boezio, and G. Zampa, "A 32-channel readout ASIC for X-ray spectrometry and tracking in the GAPS experiment," *IEEE Trans. Nucl. Sci.*, vol. 71, no. 1, pp. 96–105, Jan. 2024, ISSN: 1558-1578. DOI: 10.1109/TNS.2023.3336192

[J12] M. Verzeroli, L. Gaioni, A. Galliani, L. Ghislotti, P. Lazzaroni, and V. Re, "Advancing sustainable mobility: A data acquisition system for light vehicles and active mobility," *Electronics*, vol. 13, no. 21, p. 4249, Jan. 2024, ISSN: 2079-9292. DOI: 10.3390/electronics13214249

[J13] V. Re, L. Ghislotti, P. Lazzaroni, M. Manghisoni, E. Riceputi, L. Ratti, M. Boezio, G. Zampa, and L. Fabris, "A mixed-signal processor for x-ray spectrometry and tracking in the GAPS experiment," *Nucl. Instrum. Methods Phys. Res. Sect. Accel. Spectrometers Detect. Assoc. Equip.*, vol. 1045, p. 167 617, Jan. 2023, ISSN: 0168-9002. DOI: 10.1016/j.nima.2022.167617

Conference Proceedings

Publications in international conference proceedings related to particle physics instrumentation, ASIC design, and detector systems.

[P1] L. Ghislotti, P. Lazzaroni, M. Manghisoni, L. Ratti, V. Re, and E. Riceputi, "A 65 nm CMOS Readout ASIC with Dynamic Signal Compression for Balloon-Borne Spectrometry Applications," in *Proceedings of SIE 2025*, P. Pavan and S. Campopiano, Eds., Cham: Springer Nature Switzerland, 2026, pp. 157–164, ISBN: 978-3-032-17305-8. DOI: 10.1007/978-3-032-17305-8_16

[P2] K. Aoyama and on behalf of the GAPS Collaboration, "GAPS Detector Cooling System: Results from Antarctic Ground Tests," in *Proceedings of 39th International Cosmic Ray Conference — PoS(ICRC2025)*, vol. 501, SISSA Medialab, Sep. 2025, p. 466. DOI: 10.22323/1.501.0466

[P3] M. Manghisoni, L. Ghislotti, P. Lazzaroni, L. Ratti, V. Re, and E. Riceputi, "DTMOS Feedback Network for Dynamic Signal Compression in Low-Noise Charge Amplifier," in *2025 IEEE Nuclear Science Symposium (NSS), Medical Imaging Conference (MIC) and Room Temperature Semiconductor Detector Conference (RTSD)*, Nov. 2025, pp. 1–1. DOI: 10.1109/NSS/MIC/RTSD57106.2025.11286312

[P4] A. Stoessl and on behalf of the GAPS Collaboration, "Estimating the performance of the GAPS detector with muon ground testing data from Antarctica," in *Proceedings of 39th International Cosmic Ray Conference — PoS(ICRC2025)*, vol. 501, SISSA Medialab, Dec. 2025, p. 531. DOI: 10.22323/1.501.0531

[P5] A. Tiberio for the GAPS Collaboration, "Indirect search for dark matter with cosmic-ray antinuclei: The GAPS experiment," *J. Phys.: Conf. Ser.*, vol. 3053, no. 1, p. 012 002, Jul. 2025, ISSN: 1742-6596. DOI: 10.1088/1742-6596/3053/1/012002

[P6] K. Yee and on behalf of the GAPS Collaboration, "Antinuclei Signatures of Dark Matter and the GAPS Experiment," in *Proceedings of 39th International Cosmic Ray Conference — PoS(ICRC2025)*, vol. 501, SISSA Medialab, Sep. 2025, p. 539. DOI: 10.22323/1.501.0539

[P7] S. N. Feldman et al., "The GAPS time-of-flight detector," in *Proceedings of 38th International Cosmic Ray Conference — PoS(ICRC2023)*, vol. 444, SISSA Medialab, Sep. 2024, p. 120. DOI: 10.22323/1.444.0120

[P8] L. Ghislotti, P. Lazzaroni, M. Manghisoni, and E. Riceputi, "Low-noise wide dynamic range charge sensitive amplifier in 65 nm CMOS technology for the second flight of the GAPS experiment," in *Proc. 19th Conf. Ph.D. Res. Microelectron. Electron. (PRIME), 2024*, Jun. 2024, pp. 1–4. DOI: 10.1109/PRIME61930.2024.10559720

[P9] R. Munini, A. Lenni, and on behalf of the GAPS Collaboration, "The identification of the cosmic-ray light nuclei with the GAPS experiment," in *Proceedings of 38th International Cosmic Ray Conference — PoS(ICRC2023)*, vol. 444, SISSA Medialab, Sep. 2024, p. 179. DOI: 10.22323/1.444.0179

[P10] R. Munini and on behalf of the GAPS Collaboration, "Integration and Calibration of the GAPS Antarctic Balloon Payload," in *Proceedings of 38th International Cosmic Ray Conference — PoS(ICRC2023)*, vol. 444, SISSA Medialab, Sep. 2024, p. 180. DOI: 10.22323/1.444.0180

[P11] A. Stoessl and on behalf of the GAPS Collaboration, "The GAPS experiment - a search for light cosmic ray antinuclei," in *Proceedings of 38th International Cosmic Ray Conference — PoS(ICRC2023)*, vol. 444, SISSA Medialab, Sep. 2024, p. 1440. DOI: 10.22323/1.444.1440

[P12] M. Verzeroli, A. Galliani, L. Ghislotti, L. Gaioni, P. Lazzaroni, and V. Re, "Empowering Smart Mobility with a Component-based Data Acquisition System for Multi-sensor Readout," in *2024 19th Conference on Ph.D Research in Microelectronics and Electronics (PRIME)*, Jun. 2024, pp. 1–4. DOI: 10.1109/PRIME61930.2024.10559671

[P13] T. Aramaki et al., "GAPS contributions to the 38th International Cosmic Ray Conference (Nagoya 2023)," 2023. DOI: 10.48550/ARXIV.2310.10181

[P14] T. Aramaki et al., *GAPS contributions to the 38th international cosmic ray conference (nagoya 2023)*, Oct. 2023. DOI: 10.48550/arXiv.2310.10181 arXiv: 2310.10181 [astro-ph].

[P15] E. Riceputi, M. Boezio, L. Fabris, L. Ghislotti, P. Lazzaroni, M. Manghisoni, L. Ratti, V. Re, and G. Zampa, "The 32 Analog Channels Readout for the Long-Flight GAPS Balloon Experiment Tracking System," in *Proceedings of SIE 2022*, G. Cocorullo, F. Crupi, and E. Limiti, Eds., Cham: Springer Nature Switzerland, 2023, pp. 27–32, ISBN: 978-3-031-26066-7. DOI: 10.1007/978-3-031-26066-7_5

[P16] E. Riceputi et al., "Experimental results from the characterization of a 32-channels mixed-signal processor for the GAPS experiment," in *2023 IEEE Nuclear Science Symposium, Medical Imaging Conference and International Symposium on Room-Temperature Semiconductor Detectors (NSS MIC RTSD)*, Vancouver, BC, Canada: IEEE, Nov. 2023, pp. 1–1, ISBN: 979-8-3503-3866-9. DOI: 10.1109/NSSMICRTSD49126.2023.10338321

Conference Presentations

Oral presentations delivered at international conferences and scientific meetings in the fields of microelectronics, particle physics instrumentation, and detector readout systems.

2026 *Design and Experimental Characterization of a 65 nm CMOS Readout Channel for the Si(Li) Tracker Upgrade of the GAPS Experiment*. **L. Ghislotti**, P. Lazzaroni, M. Manghisoni, L. Ratti, V. Re, E. Riceputi. IEEE Nuclear Science Symposium, Medical Imaging Conference and Room-Temperature Semiconductor Detector Workshop (IEEE NSS, MIC and RTSD), Analog and Digital Circuits session, Granada, Spain, November 11. [Conference]

2026 *Characterization of a 65 nm CMOS Readout ASIC with Dynamic Signal Compression for Balloon-Borne Spectrometry Applications*. **L. Ghislotti**, P. Lazzaroni, M. Manghisoni, L. Ratti, V. Re, E. Riceputi. LVII Annual Meeting of the Italian Society of Electronics (SIE), L'Aquila, Italy, 2026. [Conference]

2025 *DTMOS Feedback Network for Dynamic Signal Compression in Low-Noise Charge Amplifier*. M. Manghisoni, **L. Ghislotti**, P. Lazzaroni, L. Ratti, V. Re, E. Riceputi. IEEE NSS, MIC and RTSD, 2025. [Session]

2025 *A 65 nm CMOS Readout ASIC with DTMOS CSA Feedback for Dynamic Signal Compression in Balloon-Borne Spectrometry Applications*. **L. Ghislotti**, P. Lazzaroni, M. Manghisoni, L. Ratti, V. Re, E. Riceputi. LVI Annual Meeting of the Italian Society of Electronics (SIE), Sensors, Microsystems and Instrumentation session, University of Naples "Parthenope", Naples, Italy, June 26. [Programme]

2024 *Low-Noise Wide Dynamic Range Charge Sensitive Amplifier in 65 nm CMOS Technology for the Second Flight of the GAPS Experiment*. **L. Ghislotti**, P. Lazzaroni, M. Manghisoni, E. Riceputi. PRIME 2024, Photon and Particle Detection Circuits session, Larnaca, Cyprus, June 10. [Conference]

2023 *Experimental results from the characterization of a 32 channels mixed signal processor for the GAPS experiment*. E. Riceputi, M. Boezio, L. Fabris, E. Cavazzuti, **L. Ghislotti**, P. Lazzaroni, M. Manghisoni, V. Re, L. Ratti, V. Vagelli, M. Xiao, G. Zampa. IEEE NSS, MIC and RTSD, Vancouver, Canada, November 9. [Session]

2022 *A 32-channels mixed-signal processor for the tracking system of the GAPS experiment*. V. Re, **L. Ghislotti**, P. Lazzaroni, M. Manghisoni, E. Riceputi, L. Ratti, M. Boezio, G. Zampa, L. Fabris. 15th Pisa Meeting on Advanced Detectors, La Biodola, Isola d'Elba, May 27. [PDF]

Conference Posters

Poster contributions presented at international conferences and scientific workshops in the fields of microelectronics, detector instrumentation, and astroparticle physics.

2026 *Design and Characterization of a Low-Noise Wide Dynamic Range Analog Channel in 65 nm CMOS Technology for Silicon-Strip Detector Readout*. **L. Ghislotti**, P. Lazzaroni, M. Manghisoni, L. Ratti, V. Re, E. Riceputi. TWEPP 2026, Topical Workshop on Electronics for Particle Physics, Castelldefels, Barcelona, Spain, September 28 – October 2.

2026 *AVF-MONITOR: Design and Development of a Device for Vascular Access Surveillance*. E. Riceputi, **L. Ghislotti**, S. Poloni, A. Galliani, C. E. Campiglio, A. Caroli, M. Bozzetto, L. Soliveri. LVII Annual Meeting of the Italian Society of Electronics (SIE), L'Aquila, Italy, 2026. [Conference]

2025 *The ANTARES4 Readout ASIC for the Second Flight of the GAPS Experiment*. **L. Ghislotti**, P. Lazzaroni, M. Manghisoni, E. Riceputi. Advances in Space AstroParticle Physics (ASAPP 2025), Sant Feliu de Guíxols, Girona, Spain, May 12–16. [Indico]

2024 *Design and performance of the readout chip in the Si(Li) tracker module of the GAPS experiment*. V. Re, **L. Ghislotti**, P. Lazzaroni, M. Manghisoni, E. Riceputi, L. Ratti, M. Boezio, G. Zampa, L. Fabris. 16th Pisa Meeting on Advanced Detectors, La Biodola, Isola d'Elba, May 26 – June 1. [Poster]

2023 *32-channels mixed-signal processor for the tracking system of the GAPS dark matter experiment*. E. Riceputi, M. Boezio, L. Fabris, **L. Ghislotti**, P. Lazzaroni, M. Manghisoni, L. Ratti, V. Re, G. Zampa. TWEPP 2023, Topical Workshop on Electronics for Particle Physics, Geremeas, Italy, October 1–6.

2023 *The GAPS experiment for the investigation of dark matter* (original title: *L'esperimento GAPS per l'indagine della materia oscura*). **L. Ghislotti** on behalf of the GAPS Collaboration. IFAE 2023, Incontri di Fisica delle Alte Energie, Catania, Italy, April 12–14. [Agenda]

2022 *The 32 analog channels readout for the long flight GAPS balloon experiment tracking system*. E. Riceputi, M. Boezio, L. Fabris, **L. Ghislotti**, P. Lazzaroni, M. Manghisoni, L. Ratti, V. Re, G. Zampa. LIII Annual Meeting of the Italian Society of Electronics (SIE), Pizzo, Italy, September 7–9.

Academic Activity

Teaching Activity

Since 2022, Luca Ghislotti has carried out teaching assistant activities for the degree programs in *Engineering and Management for Health*, *Mechatronics and Smart Technology Engineering*, and *Computer Engineering* at the University of Bergamo. A detailed list by academic year follows:

2022–2023
- Biomedical Sensors (6 CFU), M.Sc. in Engineering and Management for Health.
- Sensors (3 CFU), M.Sc. in Mechatronics and Smart Technology Engineering.
- Industrial IoT (3 CFU), M.Sc. in Mechatronics and Smart Technology Engineering.

2023–2024
- Biomedical Sensors (6 CFU), M.Sc. in Engineering and Management for Health. Included the seminar *"Tiny Technologies, Big Impact: Lab-on-Chip Devices, Microfluidics, and MEMS in Biomedical Engineering"*.
- Sensors (3 CFU), M.Sc. in Mechatronics and Smart Technology Engineering.
- Industrial IoT (3 CFU), M.Sc. in Mechatronics and Smart Technology Engineering.

2024–2025
- Biomedical Sensors (6 CFU), M.Sc. in Engineering and Management for Health.
- Sensors (3 CFU), M.Sc. in Mechatronics and Smart Technology Engineering.
- Smart Sensors and Electronic Systems (3 CFU), M.Sc. in Mechatronics and Smart Technology Engineering.
- Industrial IoT (3 CFU), M.Sc. in Mechatronics and Smart Technology Engineering.
- Electronic Systems Design (*Progettazione dei Sistemi Elettronici*, 6 CFU), M.Sc. in Computer Engineering. Delivered introductory lessons on PCB design with Altium Designer 25 (schematics, component libraries, PCB layout, DRC, manufacturing files).

2025–2026
- Smart Sensors and Electronic Systems (3 CFU), M.Sc. in Mechatronics and Smart Technology Engineering.
- Industrial Electronics (*Elettronica Industriale*, 6 CFU), B.Sc. in Computer Engineering.
- Biomedical Sensors (6 CFU), M.Sc. in Engineering and Management for Health.

Thesis Supervising Activity

During his Ph.D. studies, Luca Ghislotti has served as co-supervisor or reviewer for the following theses at the Department of Engineering and Applied Sciences, University of Bergamo:

- *"Characterization and calibration of a rad-hard bandgap voltage reference designed for high-energy physics applications"*, M.Sc. in Computer Engineering, A.Y. 2022/2023.

- *"Development of a MicroPython firmware in mbed-flash mode for a wearable platform"*, B.Sc. in Computer Engineering, A.Y. 2022/2023.
- *"Design and characterization of readout channels for SiPMs used in the detection of cosmic-ray muons"* (original title: *"Progetto e caratterizzazione di canali di lettura per SiPM impiegati nella rivelazione di muoni da radiazioni cosmiche"*), B.Sc. in Computer Engineering, A.Y. 2023/2024.
- *"Design and development of a low-power embedded system for the monitoring of arteriovenous fistulas"* (original title: *"Progettazione e sviluppo di un sistema embedded low-power per il monitoraggio di fistole artero-venose"*), B.Sc. in Computer Engineering, A.Y. 2024/2025.
- *"Design of an 8-bit digital-to-analog converter in a 65 nm CMOS technology for the second flight of the GAPS experiment"*, M.Sc. in Mechatronics and Smart Technology Engineering, A.Y. 2025/2026.

Research Training and Service

Visiting Periods

- **December 2022 & March 2023 — Space Sciences Laboratory (SSL), University of California, Berkeley (CA)**
  Contributed to the assembly and calibration of the GAPS tracker modules. These visits were focused on hands-on integration of the readout electronics with the silicon detector layers, including noise performance validation and threshold tuning.
- **June 2023 — National Technical Systems (NTS), Los Angeles (CA)**
  Participated in the Thermo-Vacuum (TVAC) qualification test of the GAPS tracker. In this context, worked on the thermal system of the instrument and collaborated directly with engineers from the Japan Aerospace Exploration Agency (JAXA) to validate the thermal performance under simulated flight conditions.
- **August 2023 — Pupin Physics Department, Columbia University, New York (NY)**
  Attended the GAPS Collaboration Meeting, contributing to technical discussions on tracker performance, calibration strategy, and the preparation for integration activities.
- **January & May 2024 — Nevis Laboratories, Columbia University, Irvington (NY)**
  Engaged in the assembly and system-level testing of the GAPS payload. Activities included functional verification of the Si(Li) tracker electronics, DAQ integration, and support to calibration campaigns with cosmic-ray muons.
- **May 2024 — NASA Columbia Scientific Balloon Facility (CSBF), Palestine (TX)**
  Participated in the full payload integration and testing at the CSBF facility. Conducted a dedicated Thermo-Vacuum (TVAC) validation test of the instrument power supply system and contributed to pre-flight checks in preparation for long-duration balloon operations.

- **November 2024 — NASA Long Duration Balloon (LDB) facility, McMurdo Station, Antarctica**
  Selected and qualified to take part in the Antarctic launch campaign under the NSF United States Antarctic Program (USAP). Responsibilities included integration and testing of the GAPS payload at the LDB site, coordination of tracker operations, and direct support to launch activities of the long-duration balloon mission.

Ph.D. Schools and Courses

During his Ph.D. studies, Luca Ghislotti attended the following specialized schools and training courses in the field of high-energy physics detectors and microelectronics:

- *Giornate di Studio sui Rivelatori — Scuola F. Bonaudi e E. Chiavassa*, June 26–30, 2023, Sala del Palazzo Comunale, Cogne (AO), Italy. [gsr.to.infn.it]
- *EURIZON Detector School 2023*, July 17–28, 2023, Bergische Universität Wuppertal, Germany. Delivered the student talk *"The GAPS experiment for the indirect search of dark matter"* during the session *Making engaging scientific presentations*, organized by Lucie Linssen (CERN) and David Barney (CERN). [indico.cern.ch]
- *SIE Ph.D. School 2023*, September 4–7, 2023, Università degli Studi di Messina, Italy. Organized by SIE (Società Italiana di Elettronica).
- *SystemVerilog for Design*, March 18–21, 2024, Microelectronics Support Centre, UKRI STFC Rutherford Appleton Laboratory, Harwell Campus, Didcot, UK. Organized by EUROPRACTICE. [europractice.stfc.ac.uk]
- *SIE Ph.D. School 2025*, June 23–25, 2025, Università degli Studi della Campania Luigi Vanvitelli, Naples, Italy. Organized by SIE (Società Italiana di Elettronica).
- *Scuola di Misura*, February 17, 2026, Polo Tecnologico di Pavia, Italy. Training event organized by Batter Fly, Rohde & Schwarz, and Wuerth Elektronik, focused on practical and theoretical aspects of electronic measurements, DC/DC converters characterization, control loop response, and conducted/radiated EMI measurements.
- *SIE Ph.D. School 2026 — "Industrial skills in electronic and microelectronic systems"*, June 22–24, 2026, Centro Congressi Zordan, Università degli Studi dell'Aquila, L'Aquila, Italy. Organized by SIE (Società Italiana di Elettronica), in collaboration with the University of L'Aquila. [sie2026.laquilacongressi.it]
- *Front-End Electronics for Radiation Detectors*, November 7, 2026, Hotel Saray, Granada, Spain. Organized as part of IEEE NSS, MIC and RTSD 2026. Lectures delivered by Gianluigi De Geronimo (DG Circuits, USA), covering low-noise front-end electronics design, and Lodovico Ratti (University of Pavia and INFN), covering radiation tolerance. [IEEE NSS programme]

Paper Reviewing Activity

Luca Ghislotti serves as a technical reviewer in the fields of microelectronics and electronics design for the following international journals and conferences:

International Journals

- *IEEE Transactions on Nuclear Science (TNS)*, since 2025.
  - Verified peer-review activity recorded on the Web of Science Reviewer Recognition Service (formerly Publons).
  - Reviewer Profile IDs: CtrzGmDC; V1zhxryA.

International Conferences

- *International Conference on Modern Circuits and Systems Technologies (MOCAST)*, Athens, Greece (2023) and Sofia, Bulgaria (2024).
- *International Conference on Applications in Electronics Pervading Industry, Environment and Society (ApplePies)*, Genova, Italy (2023) and Turin, Italy (2024).

Public Engagement and Outreach Activities

- **Invited Talk at the 30th Anniversary of Centro Studi Leonardo da Vinci**
  Bergamo (May 9, 2025). Invited talk to students and faculty during the school's 30th anniversary. Shared academic and professional journey from Leonardo da Vinci studies to international collaborations, emphasizing the importance of technical education, following one's passions, and his Antarctic mission with the GAPS experiment.

- **Talk: "Elettronica ai confini del mondo: tra i ghiacci dell'Antartide alla ricerca della materia oscura" (Electronics at the edge of the world: searching for dark matter among the ice of Antarctica)**
  Università degli Studi di Bergamo Open Day, presentation of the course *Ingegneria delle Tecnologie per l'Elettronica e l'Automazione* (Electronics and Automation Technology Engineering) (March 22, 2025). Talk focused on electronics for extreme environments, the GAPS experiment, and the Antarctic mission experience.

- **Seminar: "L'indagine della materia oscura con l'esperimento GAPS" (The search for dark matter with the GAPS experiment)**
  Museo di Scienze Naturali "E. Caffi", Bergamo (December 14, 2024). Public seminar delivered during the University of Bergamo event *"Voci dall'Antartide"* (Voices from Antarctica), part of the initiative *"Chiedilo all'acqua – Dialoghi sul valore di una risorsa vitale e sul suo futuro"* (Ask the water: dialogues on the value of a vital resource and its future). The meeting gathered researchers involved in Antarctic projects funded by the PNRA and in international collaborations such as the

GAPS experiment, supported by NASA, JAXA, INFN, and ASI. The event was also featured in the article *"Dal 13 al 15 dicembre: Chiedilo all'acqua, dialoghi sul valore di una risorsa vitale e sul suo futuro"* on *Montagne e Paesi* [Link].

- **Article Appearance: Corriere della Sera**
  Featured in the article *"L'acqua da indagare – Missione in Antartide"* (Water to investigate: mission to Antarctica), published on December 14, 2024 (Bergamo edition). The article highlighted Luca Ghislotti's role in the GAPS experiment and his participation in the Antarctic balloon launch campaign.

Projects and Awards

In 2023, Luca Ghislotti participated in the Bosch Sensortec *Making Sensor Tec! Challenge* with the project *"Exhaust gas pattern analysis for environmental monitoring and predictive maintenance"*. He and his team won first prize at the award ceremony held at Bosch Sensortec's headquarters in Milan (April 14, 2023) and were subsequently invited to present the project at Bosch Sensortec's main headquarters in Reutlingen, Germany (July 4, 2023).

Pursuant to art. 46 and 47 of Presidential Decree 445/2000, I declare that the information included in my CV is true, being aware of the possible application of Article 76 of the same decree in the event of a false declaration.
Pursuant to Legislative Decree (D.Lgs.) no. 196/2003 and Regulation (UE) 2016/679, the undersigned acknowledges that his personal data collected here will be processed, including in electronic form, exclusively for the purpose related to this procedure and authorizes their use accordingly.

Bergamo, 14/11/2026